American Computer Science League

2020-2021 ● Contest 3: Solutions ● Elementary Division

1. Boolean Algebra

(7 ≥ 4 * 2) OR NOT (5 − 3 < 2 AND 1+ 3 ≠ 5)
(7 ≥ 8) OR NOT (2 < 2 AND 4 ≠ 5)
FALSE OR NOT (FALSE AND TRUE)
FALSE OR NOT FALSE = FALSE OR TRUE = **TRUE**

1. TRUE (A)

2. Boolean Algebra

Use the table below to evaluate NOT (A AND NOT B) OR A:

A	B	~B	A~B	~(A~B)	~(A~B)+A
0	**0**	1	0	1	1
0	**1**	0	0	1	1
1	**0**	1	1	0	1
1	**1**	0	0	1	1

Therefore, all 4 ordered pairs make the expression true.

2. 4 (E)

3. Boolean Algebra

~(A OR B) OR ~A is TRUE when ~A is TRUE
which means (0,1) and (0,0) make it TRUE.
~(~A) AND A OR B is TRUE when A OR B is TRUE
which means (1,1), (1,0), and (0,1) make it TRUE.
(0,1) is the only ordered pair that makes both expressions TRUE.

3. (0,1) (B)

4. Boolean Algebra

~A(~AB + A~B)
~A(~AB + A~B)
~A(~AB + A~B)
= (~A~AB) + (~AA~B)
= ~AB + 0
= ~AB

4. ~AB (B)

5. Boolean Algebra A. ~A + ~A~B = ~A B. ~(A + B) = ~A~B C. ~A~A~B = ~A~B D. ~AA + ~A~B = 0 + ~A~B = ~A~B Therefore, A is the only one that is not equivalent to the others.	**5.** ~A + ~A~B (A)